The elements nickel and ferrum are abundant in the formation of Nife, the inner most (core) layer of the Earth. Nife is abbreviation of Nickel and Ferrum (Iron).

5. **Who among the following postulated the concept of geographical cycle of erosion?**
   (a) A. Holmes (b) W. M. Davis
   (c) S. W. Wooldridge (d) Kober
   (e) None of the above/More than one of the above

67th B.P.S.C. (Re-Exam) (Pre)-2021

Ans. (b)

W.M. Davis (William Morris Davis) has presented the concept of geological erosion cycle. On the other hand, Walther Penck Rejected the theory of Davis and introduced the concept of Morphological Analysis.

ii. The Solar System

1. **Which of the following does not belong to the solar system?**
   (a) Asteroids (b) Comets
   (c) Planets (d) Nebula

53rd to 55th B.P.S.C. (Pre) 2011

Ans. (d)

The solar system was formed around 4.6 billion years ago. The solar system consists sun (star), cosmic dust or masses attached with each other by gravitational force, planets, dwarf Planets, natural satellite, asteroids, comets, meteoroids and cosmic dust while Nebulae is not a part of Solar System. A nebula is a giant cloud of dust and gases in space.

2. **The scientist who first discovered that the earth revolves around the sun was –**
   (a) Newton (b) Dalton
   (c) Copernicus (d) Einstein

56th to 59th B.P.S.C. (Pre) 2015

Ans. (c)

Nicolaus Copernicus was an astronomer, scientist, mathematician of Poland. In 1514, Copernicus propounded the Heliocentric Theory of the solar system in his work 'Commentariolus' (Little Commentary). Notably, Indian astronomers Varahamihira propounded the same theory around a thousand years before Copernicus in the sixth century. He mentioned that the moon revolves around the earth and the earth revolves around the sun.

3. **What is the distance between Sun and Earth?**
   (a) 93 million miles (b) 103 million miles
   (c) 83 million miles (d) More than one of the above
   (e) None of the above

B.P.S.C. School Teacher (Class 9-10) 07-12-2023

Ans. (a)

As per NASA, the distance from Earth to the Sun is 93 million miles (149 million kilometers) and as per NCERT, the solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the earth and the sun. During its revolution around the sun, the earth is farthest from the sun (152 million km). This position of the earth is called aphelion and when the earth is nearest to the Sun (147 million km), this position is called perihelion.

4. **The distance between the Earth and the sun is greater during**
   (a) Aphelion (b) Perihelion
   (c) Summer solstice (d) More than one of the above
   (e) None of the above

BPSC (Tre-3) {Class 1-5}–2024

Ans. (a)

The solar output received at the top of the atmosphere varies slightly in a year due to the variations in the distance between the Earth and the sun. During its revolution around the sun, the Earth is farthest from the sun (Approx. 152 million km) on 4th July. This position of the Earth is called aphelion. On 3rd January, the Earth is the nearest to the sun (Approx. 147 million km). This position is called perihelion. Therefore, the annual insolation received by the Earth on 3rd January is slightly more than the amount received on 4th July.

iii. The Sun

1. **When does solar eclipse occur?**
   (a) When the sun comes between the earth and the moon.
   (b) When the Earth comes between the Sun and the Moon.
   (c) When the Moon comes between the Earth and the Sun
   (d) None of the above

47th B.P.S.C. (Pre) 2005

Ans. (c)

A solar eclipse occurs when the Moon comes between Earth and the Sun, and the Moon casts a shadow over the Earth. A solar eclipse can only take place at the phase of New Moon (Amavasya), when it passes directly between the Sun and the Earth and its shadow falls upon the Earth surface.